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Vidyasagar (Sarath Chandra) Machupalli FBCS

CORE
MVB

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Joined Aug 2015
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About

Executive IT Architect, IBM Cloud | BCS Fellow, Distinguished Architect (The Open Group Certified)

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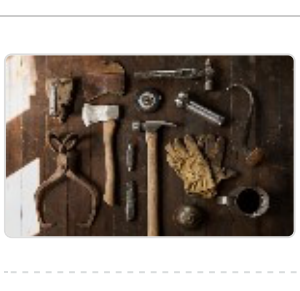

Open-Source LLM Tools Worth Your Time
Building with LLMs in 2026 means more than picking a model and calling an API. This article covers the full open-source stack by defining tools and their usage.
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Learn about when to use MCP, skills, and agents with scripts. How are they different from each other and what are they actually meant to be used for.
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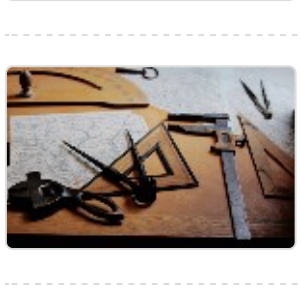

Trust No Agent: How to Secure Autonomous Tools on Your Machine
Most developers run autonomous agents with zero isolation. This guide teaches you how to build defense in depth to contain the blast radius.
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ToolOrchestra vs Mixture of Experts: Routing Intelligence at Scale
Learn about two fundamental architectural patterns reshaping how we build intelligent systems. Explore ToolOrchestra, Mixture of Experts (MoE), and other AI patterns.
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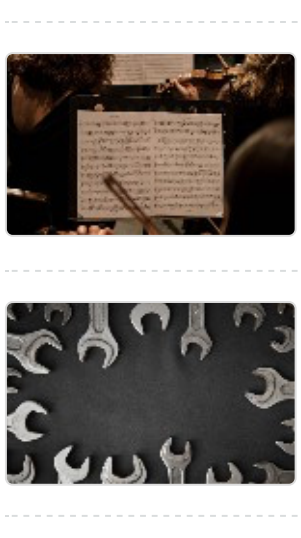

Developer Tools That Actually Matter in 2026
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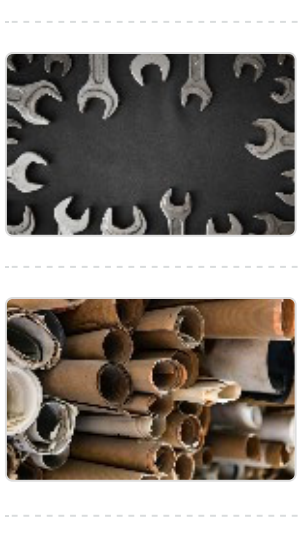

Infrastructure as Code: How Automation Evolved to Power AI Workloads
Learn about how Infrastructure as Code progressed in 2025 and how it helped automation, particularly for provisioning AI infrastructure.
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Architectural Understanding of CPUs, GPUs, and TPUs
Learn about CPUs, GPUs, and TPUs — definitions, use cases, architectural differences, and above all, when to use CPUs, GPUs, and TPUs.
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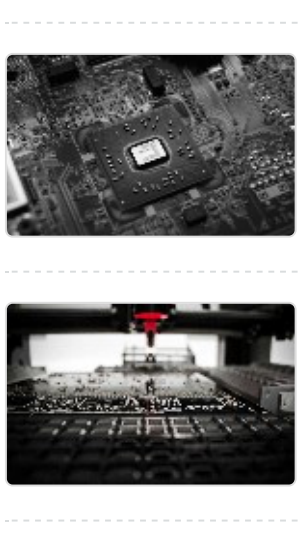

Formae and PKL: Revolutionizing Infrastructure Automation
Learn about Formae, an open-source IaC platform introducing a stateless, auto-discovering approach to infrastructure management and PKL, a configuration-as-code language.
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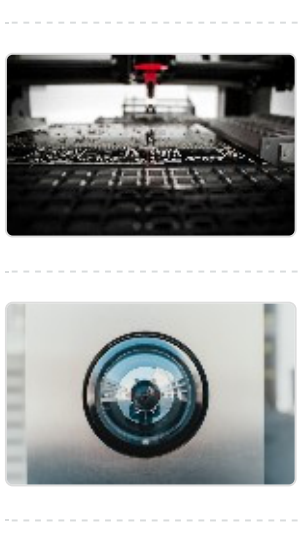

HSTS Beyond the Basics: Securing AI Infrastructure and Modern Attack Vectors
HTTP Strict Transport Security (HSTS) is a web security policy mechanism that helps protect websites against protocol downgrade attacks and cookie hijacking.
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AI Infrastructure: Compute, Storage, Observability, Security, and More
In this final article, you will learn about AI infrastructure compute, storage, observability, optimization, security, and deployment architecture.
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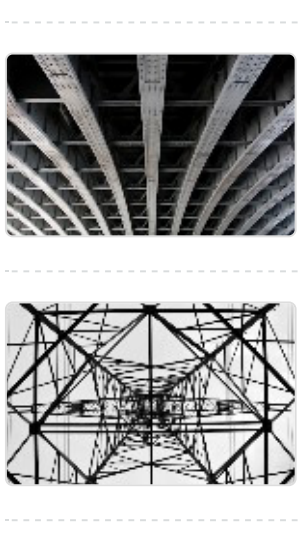

AI Infrastructure Guide: Tools, Frameworks, and Architecture Flows
This guide covers AI infrastructure, from hardware acceleration and model serving to monitoring and security, with tools, patterns, and strategies proven in production.
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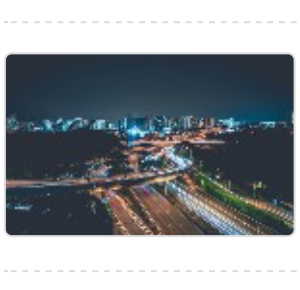

AI Infrastructure for Agents and LLMs: Options, Tools, and Optimization
This article explores the diverse infrastructure options and tools that are available for deploying and optimizing AI agents and large language models (LLMs).
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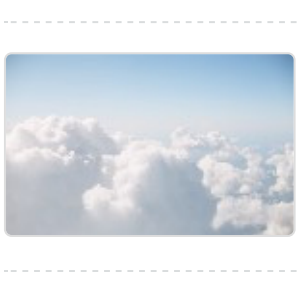

Cloud Automation Excellence: Terraform, Ansible, and Nomad for Enterprise Architecture
Enterprise cloud architecture demands sophisticated orchestration of infrastructure, configuration, and workload management across diverse computing platforms.
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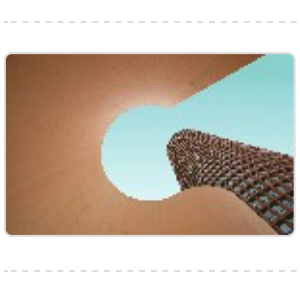

Pulumi: Modern Infrastructure as Code With Real Programming Languages
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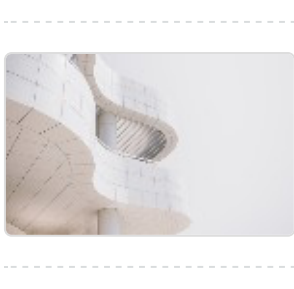

Secure Private Connectivity Between VMware and Object Storage: An Enterprise Architecture Guide
This guide details how to establish secure connectivity between VMware and Object Storage using Cloud Service Endpoints.
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DSPy Framework: A Comprehensive Technical Guide With Executable Examples
DSPy improves AI development by replacing prompt engineering with patterns. This guide explores how DSPy's features enable language model applications.
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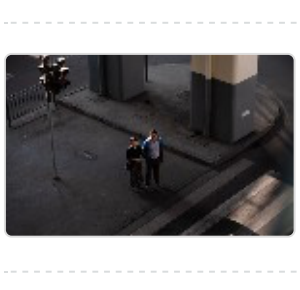

The Twelve-Factor Agents: Building Production-Ready LLM Applications
This article delves into the concept of the Twelve-Factor Agent, an architectural pattern designed to create robust, scalable, and maintainable applications.
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Advanced SSL Certificate Troubleshooting for Windows: Chain of Trust, Debugging, and Best Practices
SSL/TLS certificates are foundational to secure communications on the internet. Windows environments present unique challenges that go beyond basics.
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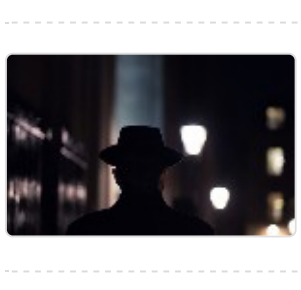

AI Agent Architectures: Patterns, Applications, and Implementation Guide
AI agent architectures provide structural blueprints for designing intelligent systems that perceive environments, process information, and execute actions.
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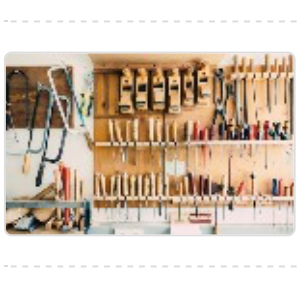

A Complete Guide to Modern AI Developer Tools
This guide explores the most impactful AI developer tools, highlighting their features, installation steps, strengths, and limitations.
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Emerging Data Architectures: The Future of Data Management
This article explores the latest advancements in data architecture, focusing on frameworks and newer paradigms such as LakeDB and zero ETL architectures.
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Ansible Security and Testing Tools for Automation
Essential collection of security and testing tools and framework for your Ansible automation. Reduce the security risk using the tools.
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Modern Data Processing Libraries: Beyond Pandas
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A Comprehensive Guide to IAM in Object Storage
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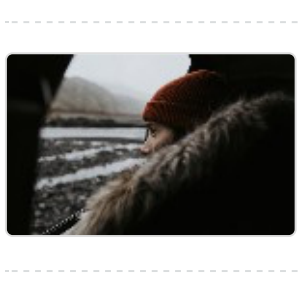

Observability and DevTool Platforms for AI Agents
Check the platforms that provide developers with powerful tools to monitor, debug, and optimize AI agents, ensuring their reliability, efficiency, and scalability.
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Logfire: Uncomplicated Observability for Python Applications
Learn about Logfire, an observability platform designed to provide developers with powerful insights into their Python applications.
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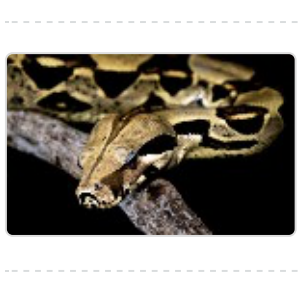

Pydantic: Simplifying Data Validation in Python
Pydantic is a powerful Python library that uses type annotations to validate data structures. Learn about the powerful features of Pydantic with code examples.
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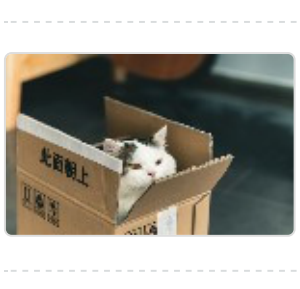

Top Tools for Object Storage and Data Management
The best tools for object storage, including MinIO, Cyberduck, and more, to efficiently manage and store unstructured data in modern cloud environments.
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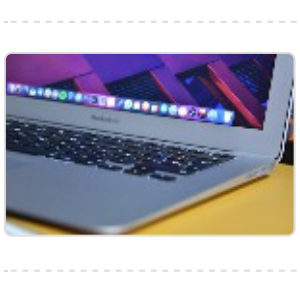

Mastering Secure Connections: A Comprehensive Guide to Accessing Sybase Databases From macOS
This article provides a step-by-step guide for configuring SSL and securely connecting to Sybase databases from macOS with tools like DataGrip.
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Data Architectures With Emphasis on Emerging Trends
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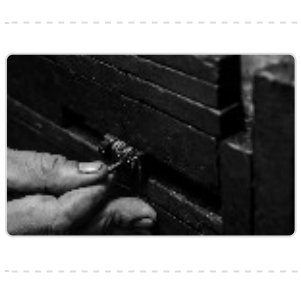

The Need for Application Security Testing
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Securing the Generative AI Frontier: Specialized Tools and Frameworks for AI Firewall
In this article, you will learn about specialized tools and frameworks for prompt inspection and protection or AI firewalls.
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Learn the essential subset of mathematical concepts to understand the underpinnings of artificial intelligence and quantum computing.
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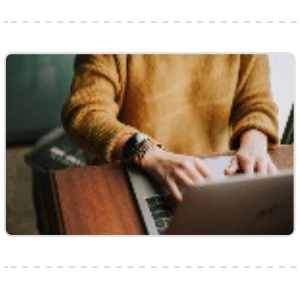

Ansible Beyond Automation
In this article, discover the wide range of tools and ecosystems that the open-source tool Ansible offers beyond automation.
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Understanding Prompt Injection and Other Risks of Generative AI
By prioritizing security, organizations can enhance trust, resilience, and reliability in their cloud and AI environments.
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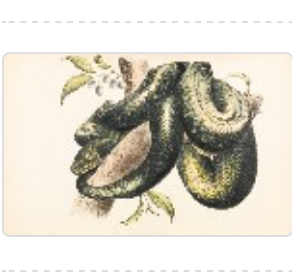

Norm of a One-Dimensional Tensor in Python Libraries
Learn how to calculate the Euclidean (norm/distance) of a single-dimensional (1D) tensor in NumPy, SciPy, Scikit-Learn, TensorFlow, and PyTorch.
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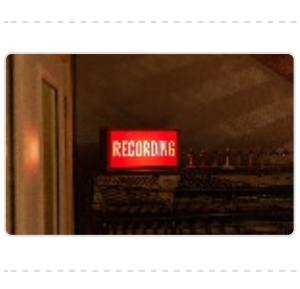

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Must-Have Infrastructure as Code Tools
A small set of tools to make your Terraform coding journey smoother. Tools to Lint, visualize, document, and clean your Terraform HCL2 code.
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Automation, Ansible, AI
Learn about how Ansible is bringing AI tools to your Integrated Development Environment to make your automation coding experience simpler, smoother, and more efficient.
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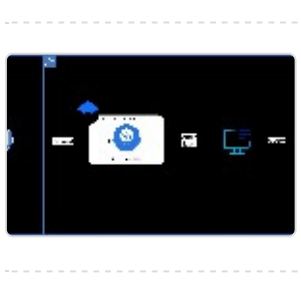

Kubernetes Auditing: Which IAM User Deleted a Namespace?
Learn how to collect audit logs that are passed through the Kubernetes API server to IBM Log Analysis to check who initiated a request and when they did so.
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Swift for C# Developers
Swift and C# are both high-level programming languages, with shared features, such as being compiled programming languages. Here's a look at Swift from the perspective of a C# developer.
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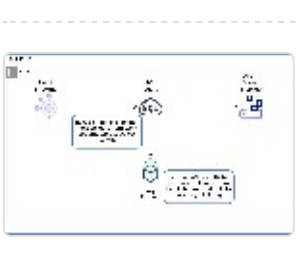

Stream Landing Kafka Data to Object Storage using Terraform
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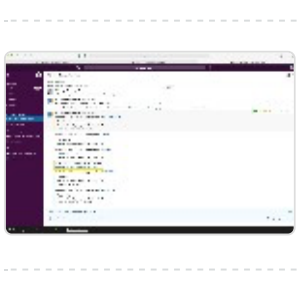

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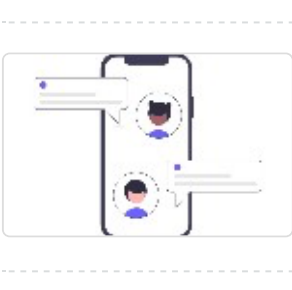

Adjust speaking rate(SSML) via cURL POST
Speaking rate is often expressed in words per minute (wpm). SSML is an XML-based markup language that provides annotations of text for speech-synthesis applications.
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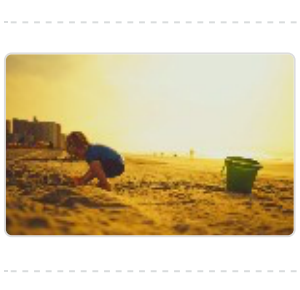

Delete a Non-Empty COS Bucket Using Terraform
Remove a Cloud Object Storage (COS) bucket that is not empty. Use a Terraform script to recursively delete all the objects of a Cloud Object Storage (COS) bucket using MinIO client.
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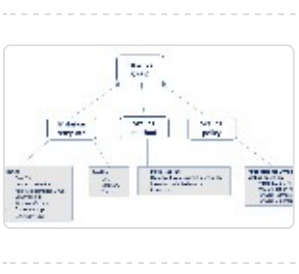

Deploy and Auto Scale Isolated Workloads Across Multiple Zones
Provision and autoscale VSI on VPC using Terraform scripts and explore autoscale scenarios like SSL termination and end-to-end encryption.
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Create Multiple Instances in a VPC Using Terraform
Learn how to provision multiple virtual server instances (VSIs) in a Virtual Private Cloud (VPC) using Terraform scripts with IBM Cloud Schematics.
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Extend VPC Instances with Cloud Functions, Activity Tracker with LogDNA, and Schematics
This post shows how to automatically assign a floating IP to a newly created VSI by monitoring Activity Tracker events and using Cloud Functions to interact with the VPC API.
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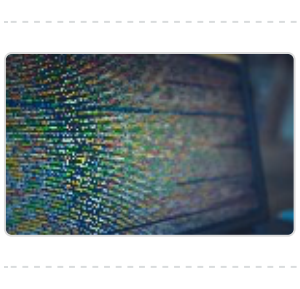

Build and Push a Container Image from Source Code With S2I
Use S2I to create a container image from your source code.
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Knative Log Analysis With LogDNA on IBM Cloud
Check out how you can track the activity going on within your cluster using IBM Log Analysis and Knative.
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Knative Monitoring with Sysdig on IBM Cloud
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Install Knative With Istio on an IKS Cluster and Deploy on IBM Cloud
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Obtain and Visualize Uniform Metrics, Logs, and Traces Across Microservices Using Istio
Learn how to set up Istio on your Kubernetes cluster and use add-ons like Grafana to collect, query, and visualize metrics and logs.
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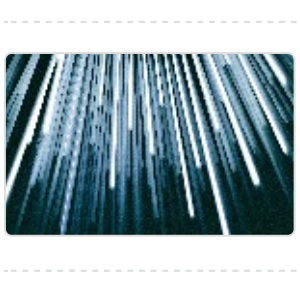

Deploy a Scalable Web Application to Kubernetes Using Helm
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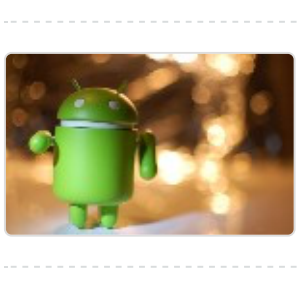

Build a Voice-Enabled Android Chatbot
Walk through the process of defining intents and entities and building a dialog flow for your chatbot to respond to customer queries.
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The Journey of a Machine Learning Model From Building to Retraining
Learn about building a predictive ML model, deploying it as an API to be used in applications, testing the model, and retraining the model with feedback data.
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Anomaly Detection in Mobile Sensor Data Using ML
This fascinating look at anomaly detection uses IoT sensors to generate data and machine learning to find unusual patterns in that data.
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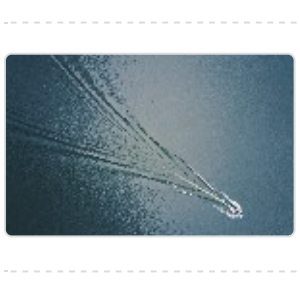

Performant Serverless Swift Actions
This tutorial goes through the steps of setting up the server side of an iOS mobile app with Swift, including the code and commands for the project.
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Personifying Chatbots: A Guide to Realistic Conversation
If you want your chatbot to pass the Turing Test, it needs to be personified and personalized. Luckily, that's not "too" difficult to do!
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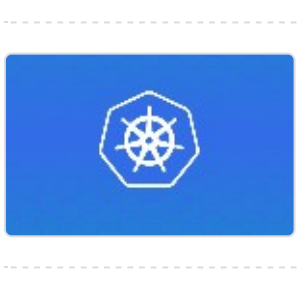
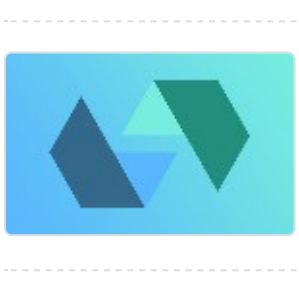
DeveloperConnect: From Containers to Serverless
Fresh from DeveloperConnect, here are a few presentations to help you get started with Bluemix, Kubernetes, and OpenWhisk.
July 13, 2017 · 2,577 Views · 4 Likes



Kubernetes From IBM Bluemix Container Service
Learn to set up clusters using Kubernetes on IBM Bluemix, expose services, and manage the logs with the following steps.
June 21, 2017 · 4,403 Views · 4 Likes

Who's Speaking?: Speaker Recognition With Watson Speech-to-Text API
Learn how to take advantage of IBM Watson's speaker diarization feature, which distinguishes between speakers in real time.
May 20, 2017 · 8,889 Views · 6 Likes

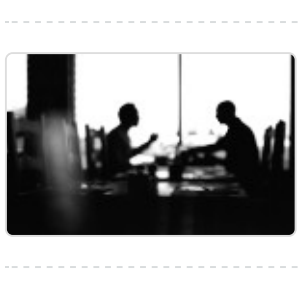
Taking a Patterns-First Approach With Bluemix [Presentation]
The Bluemix Developer Console allows devs to pick their purpose first, then start off straight away with the right tools and service integrations.
May 9, 2017 · 2,518 Views · 3 Likes



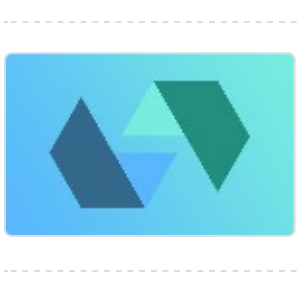
Watson Service Chaining With OpenWhisk (Part 3 of 3)
Now that you've begun chaining Watson services as an OpenWhisk sequence, it's time to expose them to the Internet as a RESTful endpoint.
April 24, 2017 · 3,324 Views · 4 Likes

Getting Containerized via the Bluemix CLI
Bluemix recently upped its Kubernetes support, but they haven't forgotten about Docker. If you like CLIs, you'll want to check out IBM Containers.
April 23, 2017 · 5,241 Views · 3 Likes

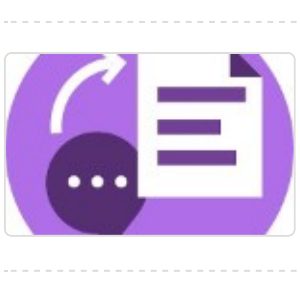
Watson Service Chaining With OpenWhisk (Part 2 of 3)
Let's get down to business. By this point, you know how serverless works and its benefits, so let's create some Watson Services and chain them together using OpenWhisk.
April 19, 2017 · 2,863 Views · 3 Likes



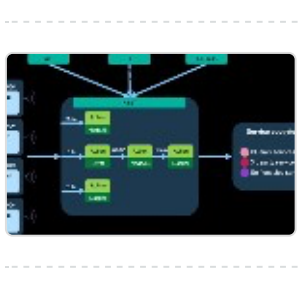
Need of Context in a Chatbot Conversation
Developers must keep in mind while building a chatbot that context is important to maintain the state of a conversation.
April 17, 2017 · 4,691 Views · 3 Likes



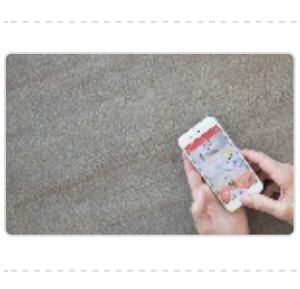
Watson Service Chaining With OpenWhisk (Part 1 of 3)
OpenWhisk can merge the power of Watson with the simple beauty of serverless computing. As we delve into Watson services, we'll cover the building blocks here.
April 17, 2017 · 4,353 Views · 5 Likes



Adding Watson Speech-to-Text to your Android App
In this post, you will learn how to create an Android application that can convert speech to text for further processing using Watson machine learning.
March 30, 2017 · 4,173 Views · 3 Likes



OpenWhisk: A FaaS Platform to Create and Run Event-Driven Apps that Scale on Demand
OpenWhisk is a powerful tool for creating both web and mobil applications. Learn how to set it up and use OpenWhisk's CLI.
March 15, 2017 · 3,596 Views · 4 Likes



Adding Watson Speech-to-Text to your Android App
Integrating AI into mobile apps is becoming more popular. Learn how to use IBM's Waston to create speech-to-text capability in your app.
March 7, 2017 · 10,183 Views · 4 Likes

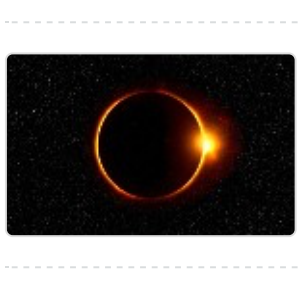
Integrating Watson Text-to-Speech Into an Android Native App
Rather than reading a message, it's always good to hear it. Integrating Watson text-to-speech into your existing Android native app lets you do just that.
Updated February 23, 2017 · 6,420 Views · 6 Likes

Walking Through Bluemix [Videos]
You can use IBM Bluemix's various cloud offerings to quickly build your apps. These videos explore how to get started with Bluemix while highlighting its ease of use.
January 15, 2017 · 3,935 Views · 2 Likes

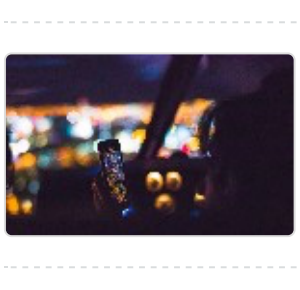
A Feedback App in Minutes With Ionic and Cloudant
Perfect for meetups and the like, you can create your own feedback app with the open source Ionic and the DBaaS Cloudant that works on iOS and Android.
December 23, 2016 · 7,245 Views · 2 Likes

Watbot: An Android Chatbot Powered by IBM Watson
Watson Conversation combines a number of cognitive techniques to help you build and train a bot – defining intents and entities and crafting dialog to simulate conversation.
December 7, 2016 · 7,368 Views · 7 Likes

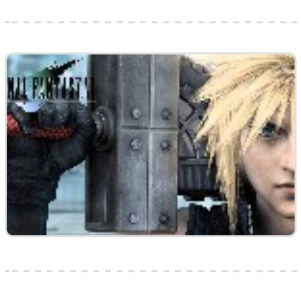
Rapidly Design, Build, Secure, and Publish an App Using Bluemix
This blog post gives you an intro to an Enhanced Bluemix Mobile Dashboard. It even has an awesome video to help you out!
December 5, 2016 · 6,288 Views · 3 Likes



MobileFirst Cordova App Development Using Eclipse
Check out Vidyasagar's tutorial information, slides, and presentation on MobileFirst Cordova app development using Eclipse.
September 15, 2016 · 4,162 Views · 7 Likes



Design, Build, and Secure an App in 12 Minutes Using Bluemix
This post guides you on how to Design, Build, Secure, and Deliver a mobile app using Mobile Dashboard on Bluemix.
August 28, 2016 · 5,733 Views · 3 Likes



IBM Cloud Tools for Swift
Swift as a programming language is now available on Linux and that means we as developers can use it on servers in data centers and in cloud. Check out some of these awesome IBM cloud tools that you should be using.
August 18, 2016 · 6,653 Views · 4 Likes

Generate Xcode Project via Swift Package Manager
Vidyasagar Machupalli didn't know how to generate an Xcode project via Swift package manager, so he figured it out and wrote a tutorial for you to learn too!
August 17, 2016 · 3,346 Views · 3 Likes

Xamarin.iOS Binding for an Existing iOS Objective-C Library with Objective Sharpie
Vidyasagar Machupalli takes us step-by-step through generating a Xamarin.iOS binding for an existing iOS Objective-C library with Objective Sharpie.
August 8, 2016 · 4,881 Views · 3 Likes

Generate Xcode Project via Swift Package Manager
Find out how to generate an Xcode project using the Swift Package Manager in this article.
July 15, 2016 · 3,890 Views · 3 Likes

IBM Cloud Tools for Swift
With Swift now available on Linux, the doors have been opened to a brave new frontier of cloud computing.
June 25, 2016 · 8,039 Views · 3 Likes

The Swift Language Explained for C# Developers
The Swift language significantly lowers the barrier to creating your own native iOS applications. Learn about its similarities to C# before you dive in.
March 2, 2016 · 5,351 Views · 7 Likes

Ionic Hybrid Mobile App Using MobileFirst Platform 7.1 CLI
Here's how to whip up an Ionic mobile app with the MobileFirst Platform, including setting up your Ionic CLI.
January 8, 2016 · 5,864 Views · 5 Likes

Integrating Xamarin Apps With IBM MobileFirst Platform
IBM MobileFirst can work quite well with Xamarin. Here's how to integrate the two, complete with an awesome video walkthrough.
January 2, 2016 · 5,085 Views · 5 Likes